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# Transformed-based foundational models in Computer Vision: an industrial use case

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**Abstract**

The adoption of deep learning models in industries for computer vision tasks is often hindered by the need to collect an adequate number of images to build a dataset sufficiently representative of the problem to be addressed. The common approach for handling this issue involves starting from a pre-trained model and finetuning it on a domain-specific dataset, drastically reducing the required number of images and, at the same time, delivering acceptable performances. Recent advances in neural network architectures such as transformers and self-supervised training techniques have contributed to the development of a particular type of deep learning model, namely, foundational models, that can be easily adapted to a wide range of downstream tasks. In this work, the performances of two foundational models were compared in a real industrial use case requiring the detection and classification of welding defects for small components. The application of these models as the backbone for a single classification linear layer trained on a small, curated dataset of 80 images, achieved an accuracy of around 95% on the test and validation sets with little implementation effort.

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**1. Introduction**

Welding is a manufacturing process that joins components, usually made of metallic or thermoplastic materials, which are fused together by means of heat. The welding process is extremely significant as almost all industrial and craft activities adopt this process. In contrast to the advancements in welding for serial production and large-scale applications, the welding of small components is characterized by fewer developments with respect to automation and control techniques. Indeed, while for the welding of large parts, which may impact workers' safety, control procedures are often defined by regulations, for small components, quality controls are usually complex and poorly standardized.

Several nondestructive testing methods, besides visual inspection, are available to detect defects or irregularities, such as Eddy Current Testing, Ultrasonic Testing, and Radiography Testing. Nevertheless, the traditional nondestructive inspections are commonly used only for large parts since they benefit from increased levels of

automation due to their scale and complexity and less frequently for small components due to high associated costs, especially when the integrity of the welding is not critical from a safety standpoint. Therefore, the control is typically limited to visual inspection by dedicated operators. However, this process is subjected to high inefficiencies and fallacies due to the subjectivity and inconsistency of the evaluation. Moreover, this inspection is related to the human visibility threshold; thus, hidden flaws may not be identified, and misjudgments could result in the delivery of defective products if no further inspections are performed.

Computer vision algorithms have recently emerged as a valuable alternative to human visual inspection, enabling the automatic analysis of images captured by cameras mounted on the production line to detect surface defects. More specifically, Convolutional Neural Network (CNN) based deep learning models have been established as the cutting-edge approach for object detection over the last ten years[1], leading to more accurate and reliable results.

CNN architectures have been considered the gold standard for computer vision tasks given their high performances, offering advanced filtering, automatic feature extraction, and spatial localization capabilities achievable by relying on a large quantity of manually labeled training data (Supervised Learning).

A deep learning model for an industrial application is commonly developed starting from a pre-trained model and finetuning it on a domain-specific dataset. The application of a pre-trained model drastically reduces the required number of training images while achieving satisfactory performances[2].

Recently, Transformer architectures[3] have demonstrated great potential to accomplish computer vision tasks, such as image classification, object detection and segmentation, and have paved the way for multimodal learning[4], which enables the analysis of different types of data simultaneously, including text, images, and audio; similar to human reasoning.

The Transformer architecture was presented for the first time in 2017[3] and has established itself as the reference architecture in the field of Natural Language Processing (NLP), given its capability to analyze data in the form of sequences, overcoming the limitations of previous models such as LTSMs[3]. Indeed, in the NLP field, task-agnostic pre-trained features are currently recognized as the standard[10] as they can be directly used without finetuning, allowing improved performances on downstream tasks compared to those attained with task-specific models.

Dosovitskiy et al.[5] demonstrated that a pure Transformer architecture is also applicable for computer vision tasks, attaining excellent results compared to the CNN gold standard, while requiring substantially fewer computational resources for training. The idea behind the so-called Vision Transformers (ViT) is to divide an image into multiple parts, called patches, and project them linearly into tokens; these tokens are analogous to those obtained from words, allowing to preserve the remaining architecture of the Transformers.

Furthermore, the Transformer architecture is also responsible for a recent paradigm shift in the AI field[6] and for the popularity of models such as BERT[7], DALL-E[8], and GPT-3[9] that can be adapted to a wide range of downstream tasks. These models have been termed “Foundational Models”[11]. A foundational model can be considered a paradigm for AI systems development wherein a model trained on large amounts of unlabeled data can be tailored to suit diverse applications. These models allow practitioners to build high-quality AI models with very few examples (few-shot learning) or no examples (zero-shot learning).

Recently, remarkable steps towards applying foundational models for computer vision have been performed thanks to research projects in self-supervised learning[12], [10], which demonstrated that it is possible to build general-purpose visual features suitable for different image distributions and tasks without finetuning. More specifically, the authors in [10], [12] implemented a self-supervised method called DINO, which can be interpreted as a form of self-distillation[13] without labeling, allowing the model to learn a well-behaved feature space by discovering object parts and shared characteristics across images. The results indicate that DINOv1 can achieve a 78.3% top-1 accuracy on ImageNet with k-NN classifiers[12]. Indeed, the possibility of using visual features without fine-tuning significantly reduces the required effort to create a suitable neural network for addressing the specific downstream task.

The purpose of this work is to test the capability of DINOv1[12] and DINOv2[10] to generate visual features exploitable in an industrial use case, without finetuning. More specifically, DINOv1 and DINOv2 were used as feature extractors to feed a single linear layer responsible for the quality classification of the welding joint in small industrial components. Experimental tests demonstrated that, using DINO features, a single linear layer can be trained following a few-shot learning approach achieving an accuracy of 95% on the test and validation sets. Therefore, the novelty of this work does not involve the introduction of new neural network architectures or new methodologies but investigates and demonstrates the possibility of exploiting foundational vision models for a particular industry application, namely welding quality classification, where datasets are highly specific to the product under analysis, and not easily standardizable.

The results illustrated in this work derive from a research study conducted in the framework of a regional-funded project: “Bando Tech Fast Lombardia - Progetto Sistemi di Visione per Piccole Saldature - VIPS” supported by Regione Lombardia, ID: 3156830. The project aims at promoting the technological and digital transformation of the production processes of SMEs in Lombardy: in particular, the project investigated the possibility of exploiting computer vision and AI algorithms for automating the quality assessment of the welding process in small components while limiting the effort required to develop and deploy the model in an industrial scenario.

In section 2 the industrial use-case is introduced, with reference to the specific industrial product under analysis; subsection 2.1 lists the tested pre-trained computer vision models and explains how they were used, whereas subsection 2.2 describes how the dataset was collected by means of a machine vision system installed on an industrial line. Section 3 illustrates the obtained results, graphically and numerically, and the employed training strategy; subsection 3.1 focuses on an additional investigation showing that it is possible to exploit DINO models for semantic segmentation. Finally, sections 4 and 5 cover respectively the discussion and the conclusions.

## 2. Material and methods

The illustrated case study focuses on a component installed in electric blankets. Electric blankets incorporate an insulated wire into the fabric that heats up when plugged in. For safety reasons, a fuse is connected to this wire, protecting the users from short circuits that might cause the blanket to overheat. The safety element (left picture of Figure 1) has an approximative dimension of 2cm(w) x 4 cm (h), whereas the electric wire cross-section is 0.1mm<sup>2</sup>. The fuse (right picture of Figure 1) is hand-soldered to the electric wire by trained workers.

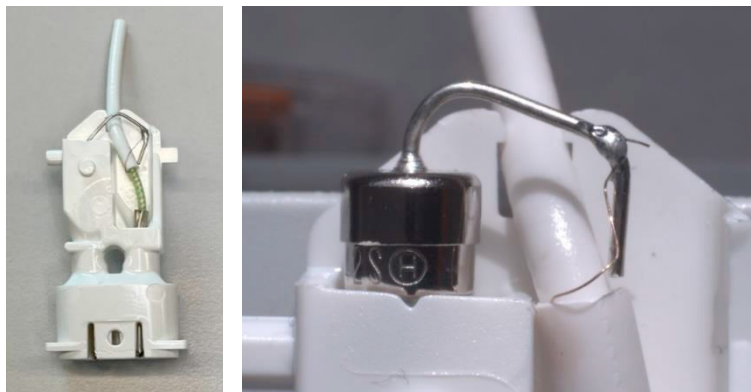


Figure 1 Safety element for electric blankets. Full view of the component (left), zoomed in on the region of the soldering with visible fuse (right)

### 2.1. Computer vision models

Several tests were performed to investigate the robustness of the features provided by two pre-trained computer vision foundational models, trained according to DINOv1[12] and DINOv2[10] methodologies. The tests were performed without modifying the network parameters, but by adding only a single linear layer for the classification task, specifically a 2-node layer.

The performances achieved with the foundational models were compared to those obtained from a well-known CNN architecture, namely ResNet[14], substituting the classification head.

The models analyzed are referenced in Table 1.

Table 1 Models evaluated during experimental tests.

| Model Architecture | Number of model parameters |
|--------------------|----------------------------|
| ViT-S/16 (DINOv1)  | 21M                        |
| ViT-S/14 (DINOv2)  | 21M                        |
| ResNet-18          | 11M                        |
| ResNet-50          | 23M                        |

## 2.2. Dataset

To build the image dataset, industrial cameras (3.2 MP, 2064x1544) equipped with 10x macro zoom lens were installed on the industrial line at a distance of about 20cm from the samples, which were illuminated by a lighting system that simulated diffused light. The machine vision system shown in Figure 2 is located right after the welding station where the operators manually solder the safety elements to the electric wire running inside the electric blankets. After soldering, the safety elements are then placed on a metal template by operators as it is possible to see in Figure 2.



Figure 2 Machine Vision system installed on the industrial line downstream the welding station.

The collected dataset consists of 200 images, 100 representing the “good” class and 100 the “defective” class, and expert operators annotated them. The whole dataset has been stratified into 80 images for training, 85 for testing, and 35 for validation. Examples of images are shown in Figure 3.

A pre-trained ResNet-18 model was used as the backbone for an 8-node linear layer to detect the weld region. The backbone parameters were kept frozen, and the linear layer was trained for the regression of the coordinates of the bounding box.

Then, the cropped images were resized, adding black borders (Figure 3) wherever needed, so that all the images had the same size of 224x224 pixels. This resolution was chosen as a multiple of 14, which is the size of the patches into which the image is decomposed into by DINO.

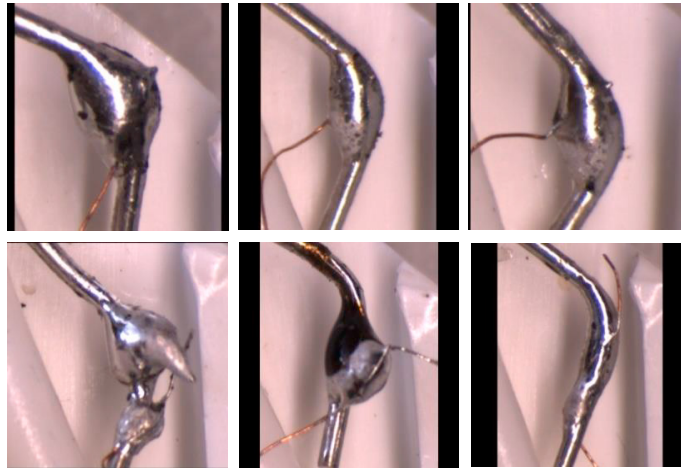


Figure 3 Cropped images used for training the classifier: good samples (top), defective samples (bottom).

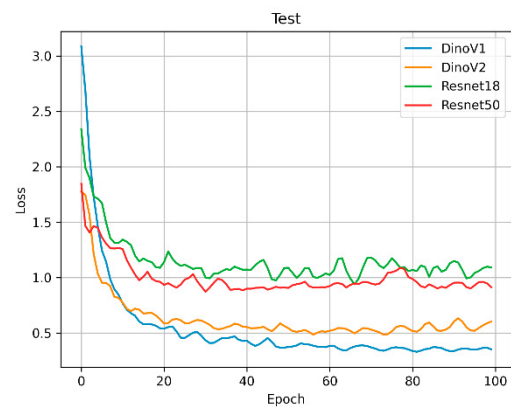
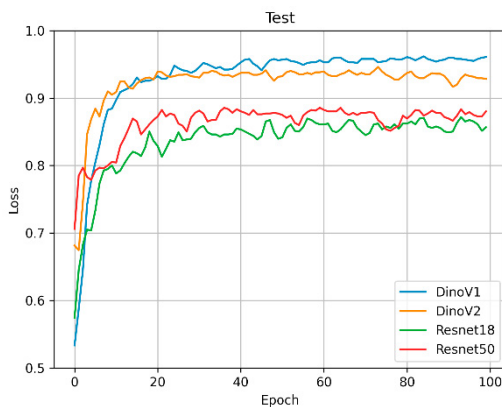
### 3. Results

The models listed in Table 1 have been applied on the cropped images to extract the features which are then provided as input to the linear classification layer to assess the welding quality: good or defective.

For the training, a batch size of 32 images was considered and 30 experiments were done, randomly creating a different train/test/validation dataset split each time. For each experiment, training was performed over 100 epochs, and, for each epoch, the models' performances were evaluated in terms of accuracy and loss (Figure 4). More specifically, the accuracy was calculated as the number of correct predictions over the number of total predictions, while the loss was evaluated using the well-known Binary Cross Entropy loss function. The optimizer used is Adam with a fixed learning rate equal to 0.001. Simple geometric transformations, such as horizontal and vertical flips or shifts, scaling, and rotational transformations, were used to augment the training dataset.

According to the accuracy graph illustrated in Figure 4, ResNet-18 and ResNet-50 behave similarly, but not as well as the DINO models, suggesting that the features extracted by the former models are equivalent when fed as input to a single linear classification layer, leading to a classification accuracy of around 85%. This result indicates a lack of robustness of these features and the need to unfreeze and finetune some backbone layers to achieve better results.

On the other hand, DINOv1 and DINOv2 demonstrated better performance than ResNet models, reaching a 95% accuracy on the test and validation set. However, neither model performed significantly better than the other.



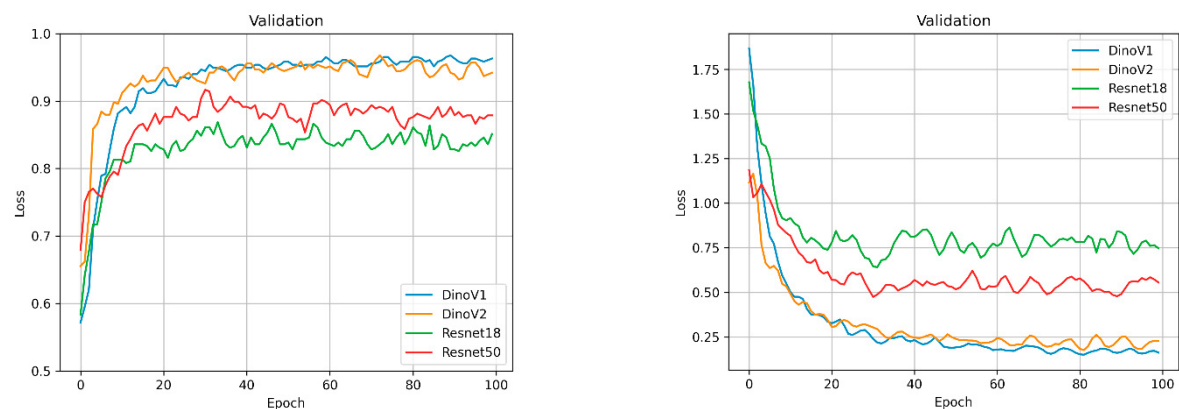


Figure 4 Accuracy (left column) and Loss (right column) for the train/test/validation datasets

The values of the graphs in Figure 4 have been averaged over the 30 different runs. In Table 2 and Table 3 the means and standard deviations of the accuracy on the validation set are reported.

Table 2 Validation accuracy mean over 30 experiments.

| Epoch | DinoV1 | DinoV2 | Resnet18 | Resnet50 |
|-------|--------|--------|----------|----------|
| 1     | 0.561  | 0.597  | 0.558    | 0.55     |
| 10    | 0.761  | 0.85   | 0.732    | 0.762    |
| 20    | 0.847  | 0.893  | 0.822    | 0.826    |
| 30    | 0.9    | 0.912  | 0.831    | 0.853    |
| 40    | 0.912  | 0.926  | 0.832    | 0.858    |
| 50    | 0.919  | 0.925  | 0.836    | 0.867    |
| 60    | 0.943  | 0.938  | 0.853    | 0.865    |
| 70    | 0.951  | 0.94   | 0.849    | 0.856    |
| 80    | 0.96   | 0.936  | 0.867    | 0.858    |
| 90    | 0.958  | 0.936  | 0.847    | 0.874    |
| 99    | 0.954  | 0.939  | 0.86     | 0.867    |

Table 3 Validation accuracy standard deviation over 30 experiments.

| Epoch | DinoV1 | DinoV2 | Resnet18 | Resnet50 |
|-------|--------|--------|----------|----------|
| 1     | 0.157  | 0.162  | 0.164    | 0.177    |
| 10    | 0.11   | 0.071  | 0.099    | 0.1      |
| 20    | 0.078  | 0.066  | 0.085    | 0.089    |
| 30    | 0.071  | 0.058  | 0.096    | 0.072    |
| 40    | 0.066  | 0.063  | 0.086    | 0.083    |
| 50    | 0.055  | 0.061  | 0.074    | 0.087    |
| 60    | 0.053  | 0.052  | 0.088    | 0.071    |
| 70    | 0.05   | 0.04   | 0.076    | 0.082    |
| 80    | 0.046  | 0.046  | 0.067    | 0.067    |
| 90    | 0.046  | 0.039  | 0.091    | 0.077    |
| 99    | 0.048  | 0.039  | 0.08     | 0.086    |

### 3.1. Features extraction considerations

An additional investigation was performed to verify that the self-supervised ViT features contain explicit information about the semantic segmentation of the images, as reported by the authors of DINO[10], [12].

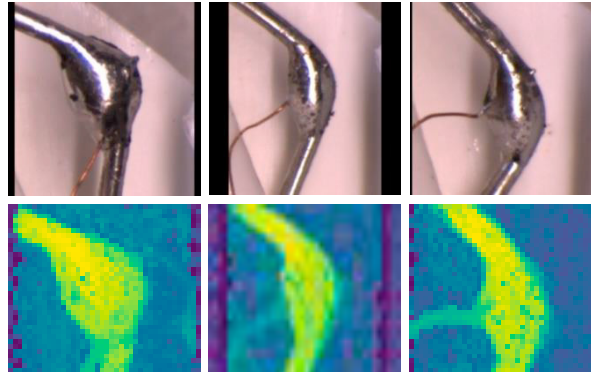


Figure 5 First PCA components of patches' features: input images (top); semantic segmentation (bottom).

More specifically, the application of the DINO models decomposes each cropped image into patches (sized e.g. 14x14 or 16x16 pixels) and extracts a vector of features (starting from 384 for the small version of ViT, the one used in the experiment, up to 1536 for the biggest ViT). To verify the above assertion, a 3-PCA was computed between the patches extracted from the test images to reduce the dimension of the features vectors down to 3 components for each patch. Then, the first component was plotted as shown in the bottom row of Figure 5. This analysis demonstrates that the features extracted by DINO can separate the soldering joint from the background without any specific finetuning, confirming that these features contain explicit information about the semantic segmentation of the images. This out-of-the-box capability of the model allows for an initial processing of the image exploitable for masking out irrelevant regions belonging to the background.

## 4. Discussion

In this work, the robustness of visual features of pre-trained models has been tested by comparing two state-of-the-art architectures, namely ViT (based on Transformers) and ResNet (based on CNNs). ViT models were pre-trained following a simple self-supervised method, namely DINO, whereas ResNets were pre-trained following a supervised methodology. The purpose of this investigation was to assess whether pre-trained models can extract features that can be used without fine-tuning for downstream tasks. For this purpose, these features were used as input for a linear layer for the classification of the quality (good vs. defective) of soldering joints in small components, exploiting a real industrial use case as a test bench.

The results demonstrated that the two tested ResNet models performed similarly, despite one having twice as many parameters as the other. In contrast, DINO pre-trained models exhibited remarkable performances and semantic segmentation capabilities out-of-the-box, further evidence that self-supervised ViT features have potential capabilities not exhibited by supervised CNNs, as pointed out by the authors of DINO. Indeed, the models relying on DINO features reached an accuracy of 95% in classifying the soldering quality by training the classification layer on only 80 images.

## 5. Conclusions

The results of this study demonstrated that computer vision foundational models, such as DINO, can provide remarkable performances while drastically reducing the effort required to adapt AI models to the industrial context.

Indeed, it was possible to quickly train and deploy an AI model to identify the weld zone to build the training dataset and to enable the company to monitor the weld quality trend qualitatively and quantitatively over time.

Additionally, new operators can immediately assess the result of their work by getting feedback from the AI system, which is valuable for deciding whether to discard the piece, and also in terms of employee training.

Future developments might involve comparing supervised and self-supervised features by feeding them to a non-linear block instead of a linear one. On one hand, this test would confirm that supervised features are the actual bottleneck and not the linear layer and, on the other hand, discover if performances could be further improved if a non-linear component is added to the pipeline.

## 6. References

- [1] S. S. A. Zaidi, M. S. Ansari, A. Aslam, N. Kanwal, M. Asghar, and B. Lee, “A survey of modern deep learning based object detection models,” *Digital Signal Processing: A Review Journal*, vol. 126. 2022. doi: 10.1016/j.dsp.2022.103514.
- [2] D. Hendrycks, K. Lee, and M. Mazeika, “Using pre-training can improve model robustness and uncertainty,” in *36th International Conference on Machine Learning, ICML 2019*, 2019.
- [3] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems*, 2017.
- [4] T. Baltrusaitis, C. Ahuja, and L. P. Morency, “Multimodal Machine Learning: A Survey and Taxonomy,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2. 2019. doi: 10.1109/TPAMI.2018.2798607.
- [5] A. Dosovitskiy et al., “AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE,” in *ICLR 2021 - 9th International Conference on Learning Representations*, 2021.
- [6] R. Bommasani et al., “On the Opportunities and Risks of Foundation Models,” *CoRR*, vol. abs/2108.07258, 2021, [Online]. Available: <https://arxiv.org/abs/2108.07258>
- [7] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *NAACL HLT 2019 - 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies - Proceedings of the Conference*, 2019.
- [8] A. Ramesh et al., “Zero-Shot Text-to-Image Generation,” *CoRR*, vol. abs/2102.12092, 2021, [Online]. Available: <https://arxiv.org/abs/2102.12092>
- [9] T. B. Brown et al., “Language Models are Few-Shot Learners,” *CoRR*, vol. abs/2005.14165, 2020, [Online]. Available: <https://arxiv.org/abs/2005.14165>
- [10] M. Oquab et al., “DINOv2: Learning Robust Visual Features without Supervision,” *ArXiv*, vol. abs/2304.07193, 2023, [Online]. Available: <https://arxiv.org/abs/2304.07193>
- [11] R. Bommasani et al., “On the Opportunities and Risks of Foundation Models,” *CoRR*, vol. abs/2108.07258, 2021, [Online]. Available: <https://arxiv.org/abs/2108.07258>
- [12] M. Caron et al., “Emerging Properties in Self-Supervised Vision Transformers,” in *Proceedings of the IEEE International Conference on Computer Vision*, 2021. doi: 10.1109/ICCV48922.2021.00951.
- [13] G. E. Hinton, O. Vinyals, and J. Dean, “Distilling the Knowledge in a Neural Network,” *ArXiv*, vol. abs/1503.02531, 2015, [Online]. Available: <https://arxiv.org/abs/1503.02531>
- [14] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2016. doi: 10.1109/CVPR.2016.90.